

BIOLOGY 413
PRINCIPLES OF BIOLOGY I
Fall, 2017

Dr. Stephen R. Pugh
Office: Rm. 557
Phone: 641-4128

E-mail: spugh@unh.edu
Office Hours: M/T/Th 11-12
and by appointment

Objectives: Upon completion of this course you should have acquired a basic understanding of biological chemistry, cell structure, cell division, genetics, and cell metabolism.

Prerequisites: None

Textbooks: Campbell, et al. 2016. *Biology in Focus*, 2nd Edition. Pearson. ISBN 978-0-321-96275-1. You may use any format you choose. You may choose to have a Mastering Biology Access Code included, because this is a very useful tool you can use to study content and prepare for exams, but it is optional. (<http://www.masteringbiology.com>).

Hofmann, A, H. 2016. *Writing in the Biological Sciences*, 2nd edition. Oxford University Press. ISBN 9780190245603.

A lab coat is required and may be purchased through the UNH-M bookstore. You do not need to purchase a lab manual.

Grading:	3 lecture exams @ 100 points.....	300 pts
	Best 10 of 13 quizzes taken @ 10 points each.....	100 pts
	Laboratory.....	125 pts
	Total Points Available.....	525 pts

All exams must be taken when scheduled. Make-up exams will be permitted only for excused absences (e.g., confirmed illness or family emergency) or if arranged in advance. Each exam only covers material since the previous exam; i.e., the second and third exams are not cumulative. Quizzes will consist of 10 multiple choice questions covering material from the previous class. They must be taken on MyCourses and must be taken by the due date. **The lowest 3 grades for quizzes that you take will be dropped** (that means if you don't take a quiz, the zero counts). There are no makeups of quizzes. Your course grade will be determined based on the percentage of points earned out of the possible 525. 90% and above will earn A- or better; 80% and above will earn B- or better; 70% and above will earn C- or better; 60% and above will earn D- or better.

Attendance:

Attendance in both lecture and laboratory is required. Excessive absences may result in a reduction of your grade.

ADA Statement:

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Disability Services Office. The Disability Services Coordinator at UNHM is Jenessa Zurek. Jenessa can be contacted at (603) 641-4383, Jenessa.Zurek@unh.edu, or in person in the Student Services Suite.

Academic Honesty:

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<https://manchester.unh.edu/current-students/academic-counseling#student-rights>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Academic Support/Tutoring:

The Center for Academic Enrichment is a resource for all students enrolled in UNH Manchester courses. Our mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. Support for other content areas is dependent on the background and competencies of current peer tutor staff. All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course. In addition, peer and professional tutors may run study groups for your class and provide drop-in math tutoring. Please visit us online at manchester.unh.edu/student/center-academic-enrichment to learn more about our services or to make an appointment with a tutor.

Lecture Schedule:

<u>Date</u>	<u>Topics</u>	<u>Reading</u>
Aug 28	Course introduction. What is biology? What is life? Scientific method.	Chapter 1
Sep 11	The chemical context of life	Chapter 2
Sep 18	The chemical diversity of life	Chapter 3
Sep 25	A tour of the cell	Chapter 4
Oct 2	Membrane transport and cell signaling	Chapter 5
Oct 9	EXAM I	
Oct 16	The cell cycle, and meiosis	Chapters 9-10
Oct 23	Inheritance, genes, and chromosomes	Chapters 11-12
Oct 30	The molecular basis of inheritance	Chapter 13
Nov 6	Gene expression: From gene to protein	Chapter 14
Nov 13	EXAM II	
Nov 20	Gene regulation, viruses, & biotechnology	Chapters 15-18
Nov 27	Introduction to metabolism and cellular respiration	Chapters 6-7
Dec 4	Cellular respiration & photosynthesis	Chapters 7-8
Dec 11	EXAM III	

BIOL 413 Laboratory

Fall 2017

Contact Information:

Keith Legro, Lab Coordinator

Ph.: (603) 641-4110

Email: keith.legro@unh.edu

Office:

Rm 550B

Feel free to contact me at any time. Email works best, I will attempt to respond to you as soon as possible.

Lab Schedule:

Module 1: Scientific Method

Aug 30, 31, Sep 1	Safety, Introduction to Brine Shrimp and Scientific Method
Sep 6, 7, 8	Experimental Design
Sep 13, 14, 15	Run Experiment, Data Collection, Statistical Analysis
Sep 20, 21, 22	Communicating in Science; Anatomy of a Journal Article, Conventions, Presentations **Quiz 1 brine shrimp
Sep 27, 28, 29	Student Presentations

Module 2: Cells

Oct 4, 5, 6	Mitosis, Meiosis and Cell Division
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Module 3: Molecular Genetics

Oct 11, 12, 13	Crime Scene Evidence Collecting & Blood Typing **Quiz 2 Mitosis/Meiosis
Oct 18, 19, 20	Crime Scene ELISA Test **Quiz 3 Blood Typing
Oct 25, 26, 27	Crime Scene Restriction Fragment Analysis ** Quiz 4 ELISA
Nov 1, 2, 3	Crime Scene Final Analysis & Communicating in Science – Forensic report ** Quiz 5 RFLP
Nov 8, 9, 10	Veterans Day No Labs
Nov 15, 16, 17	Student Presentations
Nov 22, 23, 24	Thanksgiving No Labs

Module 4: Cellular Respiration and Metabolism

Nov 29, 30, Dec 1	Metabolism
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Dec 6, 7, 8	Lab Exam
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Grading:

Lab grade will be a total of 125pts made up of the following:

Presentation/Lab Report from Module 1	20pts
Presentation/Lab Report from module 3	20pts
Lab Report from Module 2	10pts
Lab Report from module 4	10pts
Lab Quizzes	25pts
Lab Exam	20pts
Attendance	20pts
Total	125pts

Attendance:

Attendance in lab is very important as each lab of a module builds upon another. If you miss a lab you will be responsible for gathering the information needed to come to the next lab ready to perform the lab of that week. Also we will be working as groups during this lab and it is not fair to the others in your group to miss lab as that will leave the other members of the group more work to do. In light of these facts **5 pts will be deducted from your lab grade for each missed lab**. Any more than 2 missed labs will result in possible failure of the lab.

Canvas (My Courses):

All of the materials for this lab are found in your lab section of Canvas. While there may be times where you will be given hard copies of materials you will need for lab there will also be times where you will need to print out and bring the materials to lab. It is also a good way for you to review any videos or interactive programs that we may use in lab.

Electronic Devices:

You are welcome to use electronic devices (cell phones, tablets, laptops) in lab for “lab related” purposes. If you however need, or any other non-lab related use of these devices (i.e. to take or place a phone call) please be respectful of your lab mates and go outside the lab for such activities.

Lab Notebook:

You will need a notebook for lab to record your experimental progress and data in.

Lab Manual:

There is no lab manual that needs to be purchased for this lab.

To do well in this lab:

- Attend all labs! If you miss a lab you will miss an experience that cannot be replicated by simply reading the lab.
- Be prepared when coming to lab. Most material is available in advance of the lab meeting time on the Blackboard site.
- Make notes and sketches during lab. Ask questions, ask questions, ask questions!
- Fill out lab reports completely, and answer questions with more than a word or two.
- Study for the practical exam *every week* by reviewing all handouts, lab reports, notes & sketches. If you wait till just before the practical it will be too much information.